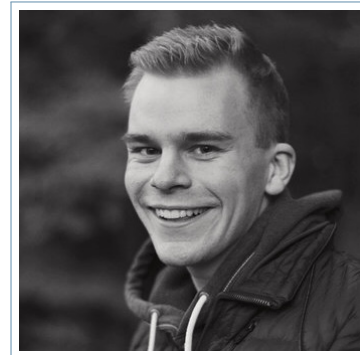


Lennart Bedarf



Personal Profile

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nationality German

date of birth 01.09.1999

University Education

- 06/2024 – present **PhD Biomedical Engineering**, *University of Basel*, Basel, Switzerland, Multi-modal Brain Tissue Analysis: From Forensic Temperature-Based Postmortem Interval Estimation to Advanced qMRI Anisotropy and Histological Validation Techniques
- 04/2022 – 01/2024 **M.Eng. Medical Engineering**, *Fachhochschule Südwestfalen (University of Applied Sciences)*, Hagen, Germany, Focus: Advanced imaging techniques in MRI
Master's thesis: "Studies on edited magnetic resonance spectroscopy: Phantom- and *in vivo*- measurements for neuroscientific problems" – grade: **1.0**
Overall grade: **1.0** (very good - with honours)
- 09/2018 – 04/2022 **B.Eng. Medical Engineering**, *Fachhochschule Südwestfalen (University of Applied Sciences)*, Hagen, Germany, Bachelor's thesis: "Python-based analysis and evaluation of spectroscopic *in vivo* MR data" – grade: **1.0**
Overall grade: **1.5** (very good)

School Education

- 2006 – 2010 **Primary school diploma**, *Grundschule Bierbaum*, Lüdenscheid, Germany
- 2010 – 2018 **General university entrance qualification (*Abitur*)**, *Zeppelin-Gymnasium*, Lüdenscheid, Germany

Work Experience

- 12/2022 – 06/2024 **Working Student**, *TEBIT GmbH & Co. KG*, Meinerzhagen, Germany
○ First real industrial experience in high precision medical engineering.
- 2020 – 2022 **COVID-19 Tester**, *Johanniter Seniorenheime GmbH*, Lüdenscheid, Germany
○ Performed rapid antigen tests for visitors prior to entering the nursing home.

Additional Qualifications

- 09/2021 – 08/2023 **Deutschlandstipendium scholarship**, *Federal Ministry of Education and Research (BMBF)*
○ Awarded to students with outstanding academic performance.
- 06/2025 **Good Scientific Practice and Its Pitfalls in Every-Day Research**, *University of Basel*, Basel, Switzerland
- 02/2025 **Presentation Training**, *University of Basel*, Basel, Switzerland
- 10/2025 **Self-Branding and Self-Promotion**, *University of Basel*, Basel, Switzerland
- 10/2025 **Stress Management in Performance Situations**, *University of Basel*, Basel, Switzerland

Skills and Interests

Languages

German Native
English Fluent
Latin Basic

Small Latinum

Software

Proficient C, Python, MATLAB, Office applications, CAD, Linux, $\text{\LaTeX}$, Git, SolidWorks

Interests

Handball Since 2004
Cooking e.g., catering for 30 people during a youth camp (one week)
Volunteering Evangelische Versöhnungskirchengemeinde Lüdenscheid
Competition “Formula 1 in Schools” international competition — successful participation

Miscellaneous

Engagement **Chairperson at the 2025 ESMRMB Annual Meeting in Marseille**
Mobility Class B driving licence

Scientific Publications

- Conference abstract **L. Bedarf**, K. Gerlach, H. Wittig, E. Scheurer, C. Lenz: "Time of Death estimation using forehead temperature measurements" In: *2026 IAFS Triannual Meeting in Sofia*, 2026.
- Paper **L. Bedarf**, K. Gerlach, H. Wittig, E. Scheurer, Celine Berger, C. Lenz: "Evaluating the forehead temperature for estimating the postmortem interval" In: *International Journal of Legal Medicine*, 2026. DOI: 10.1007/s00414-026-03736-x
- Conference abstract **L. Bedarf**, D. Neuhaus, M. J. Wendebourg, R. Schlaeger, C. Birkel, E. Scheurer, C. Lenz, : "Quantifying the Impact of Formalin Fixation on R_2^* Orientation Dependency in Human Brain White Matter [PG253]" In: *2025 ESMRMB Annual Meeting in Marseille*, 2025.
- Paper J. Bottoni, H. Wittig, T. Rost, A. Schocker, P. Wild, U. Nachbur, D. Neuhaus, **L. Bedarf**, K. Gerlach, E. Scheurer, C. Lenz: "Detection of gunshot residues using infrared photography: influence of ammunition type, surface color and blood contamination" In: *International Journal of Legal Medicine*, 2025. DOI: 10.1007/s00414-025-03609-9
- Conference abstract **L. Bedarf**, A. Holl, H. Nawrath: "Determination of T_1 and T_2 relaxation times of Phosphorus-31 in a low-field MRI by developing a Phosphorus-31 resonant low-field MRI coil." In: *Proceedings of the International Society for Magnetic Resonance in Medicine (ISMRM)*, 2023.

Basel, 01.06.2026
Lennart Bedarf